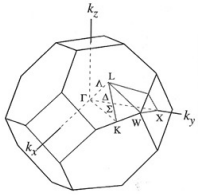
Band Structure of Semiconductors

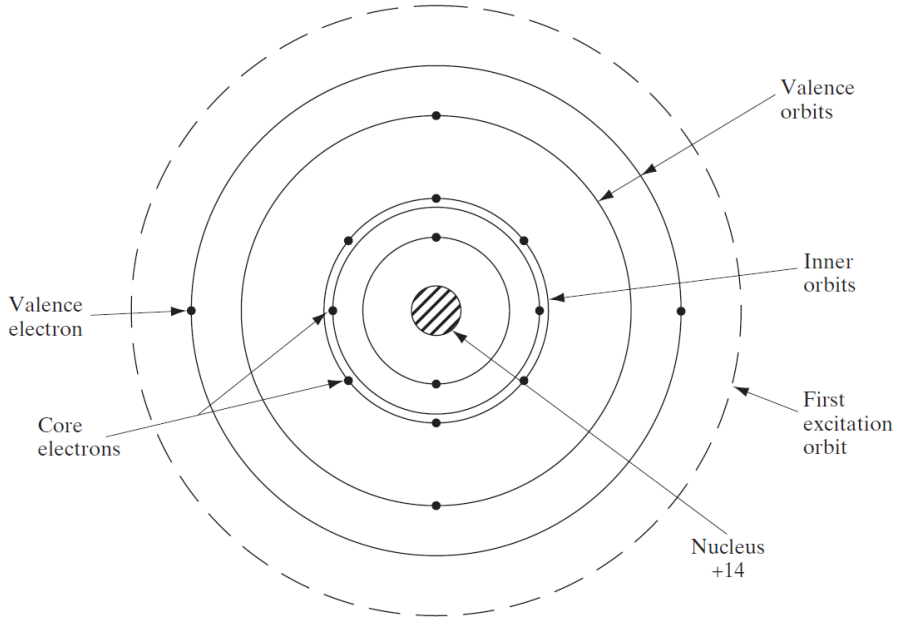
Baile Chen

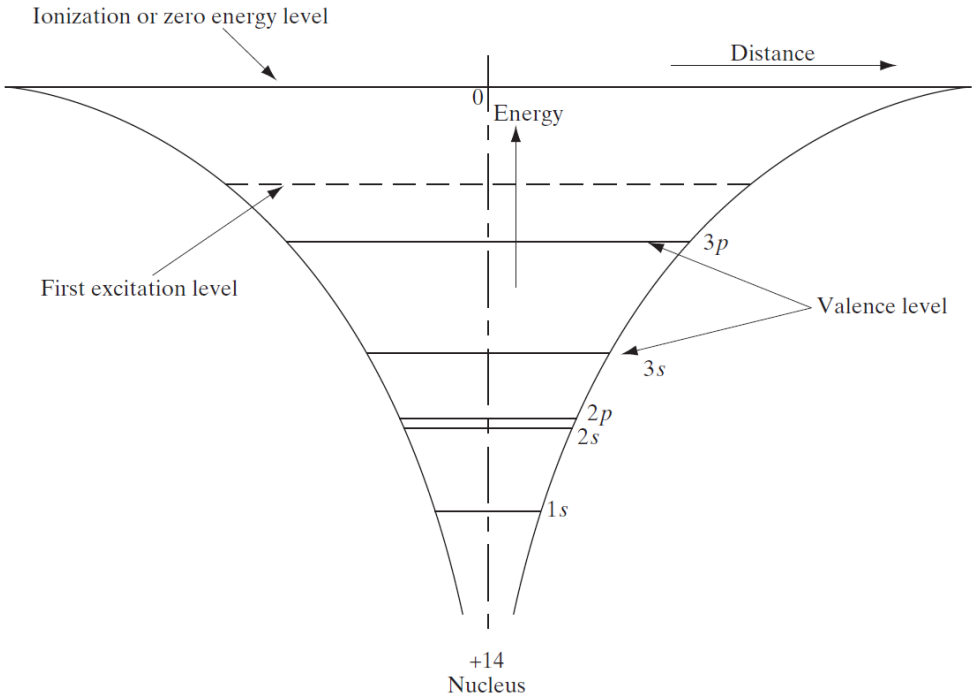
School of Information Science and Technology

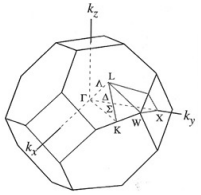
Shanghai Tech University



The Bohr Model of the (Isolated) Si Atom

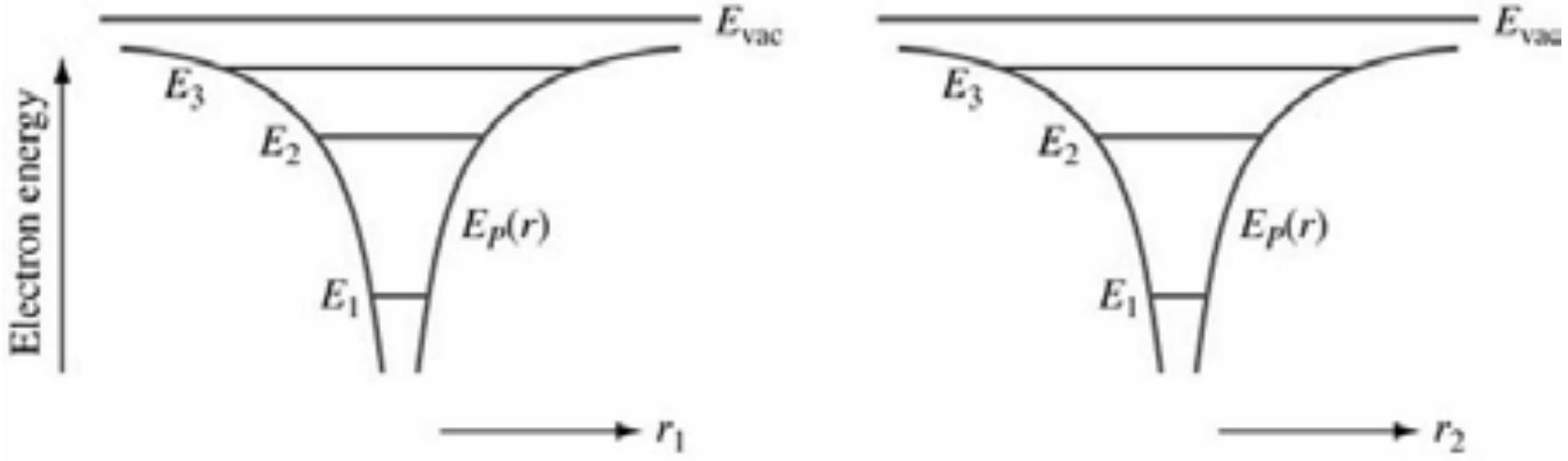


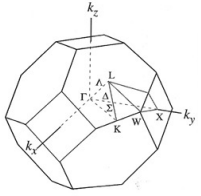




Energy Level of Two Atoms

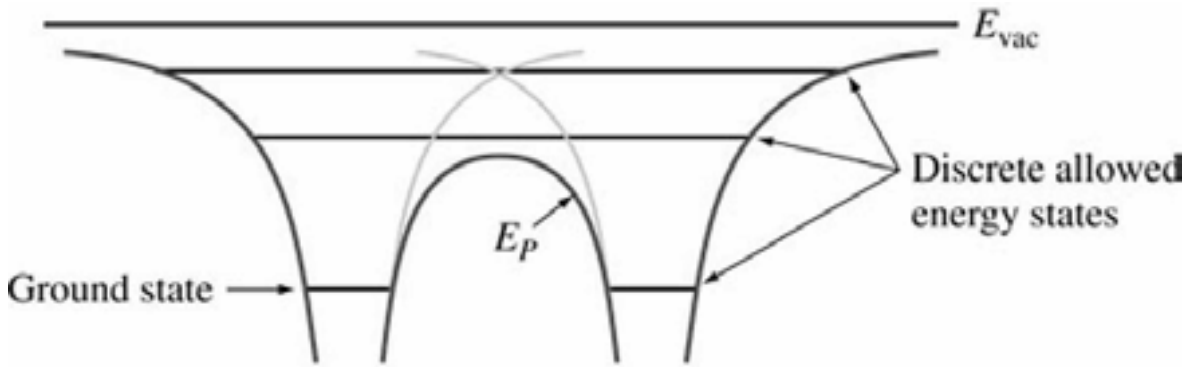
- When atoms are far apart:

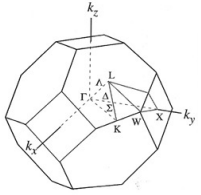




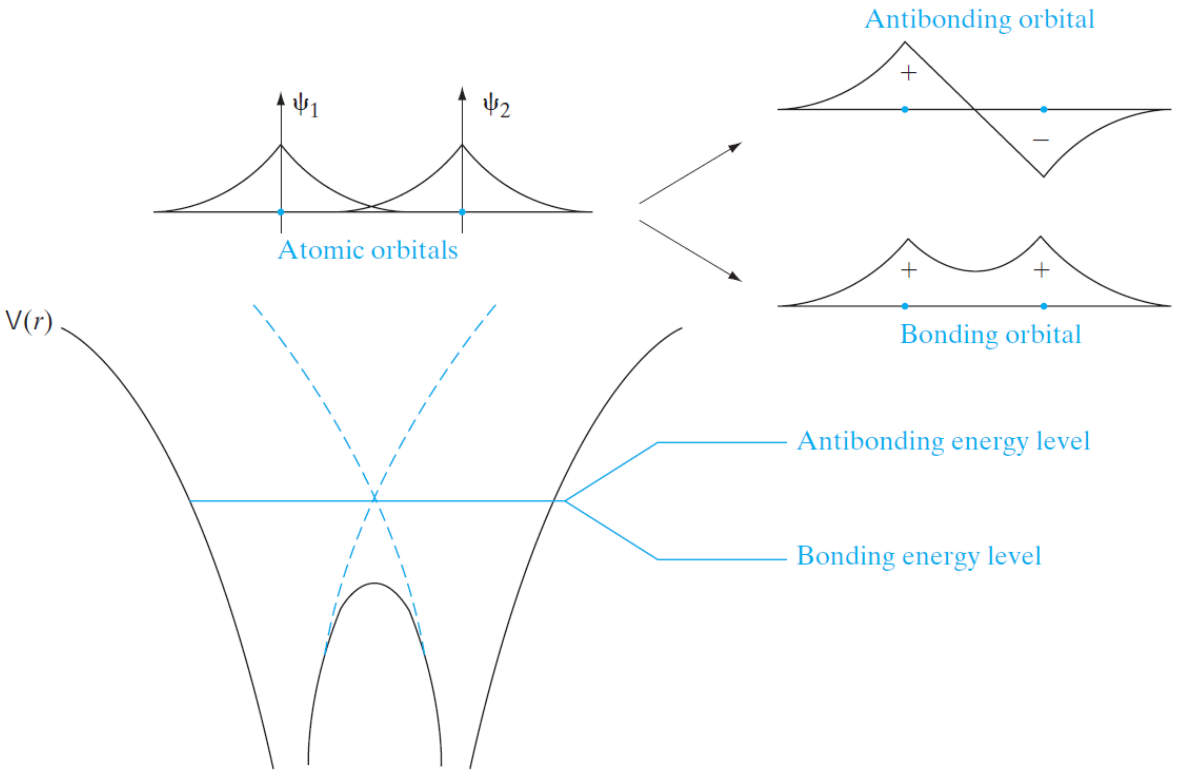
Energy Level of Two Atoms

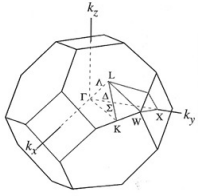
- When atoms are close together:
  - Their potential wells interact



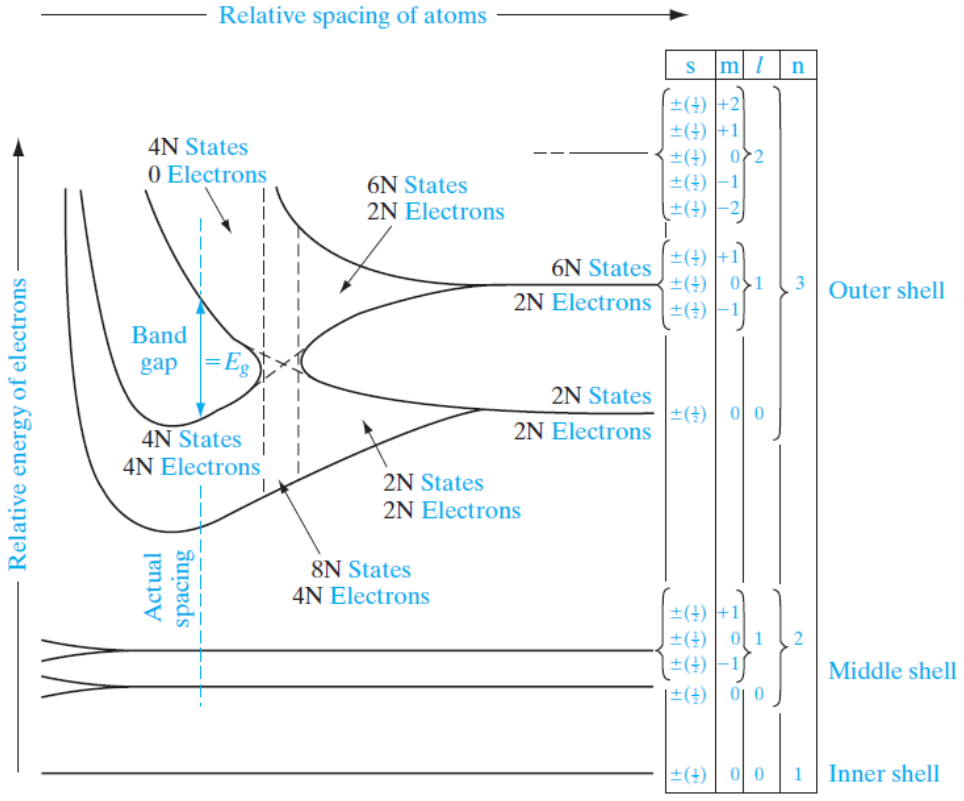


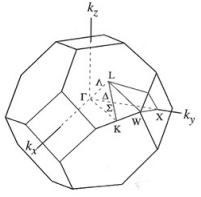
Linear Combination of Atomic Orbital (LCAO)



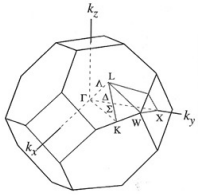


Energy Levels From Discrete Atoms to Crystal Lattice

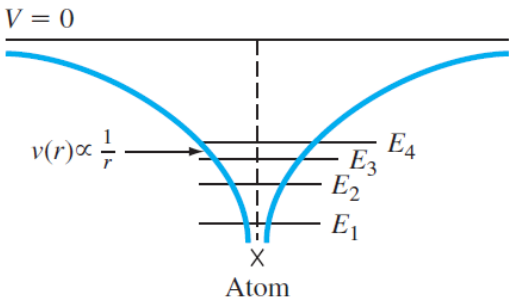


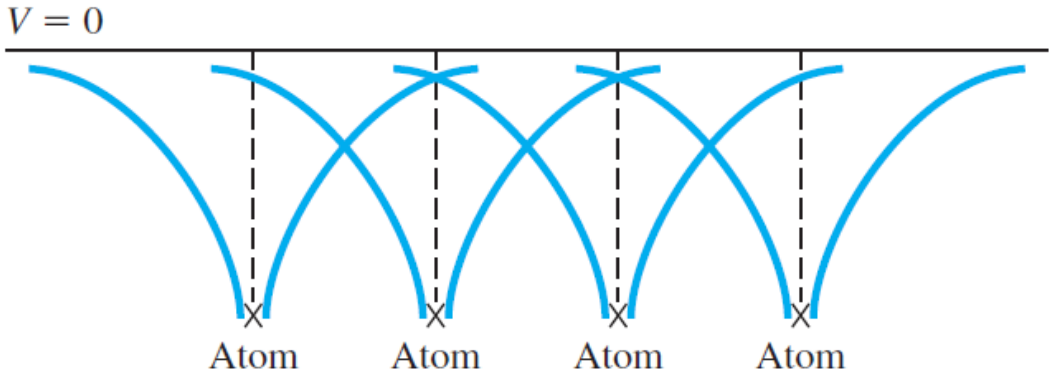


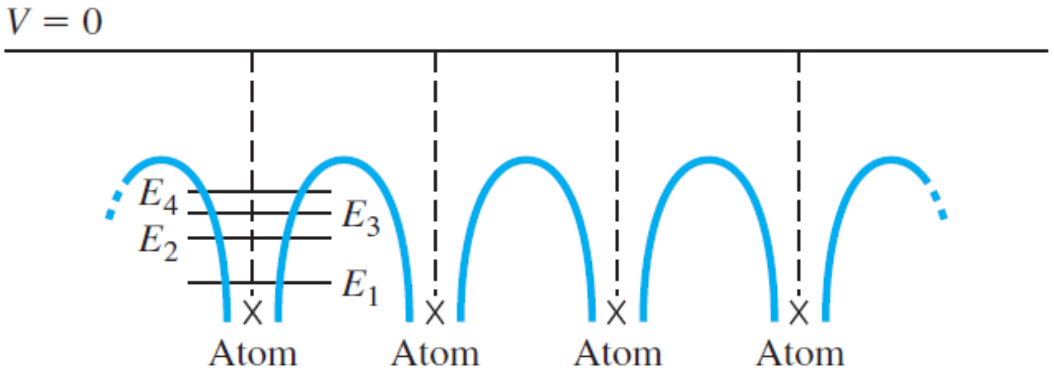
Kronig-Penny Model

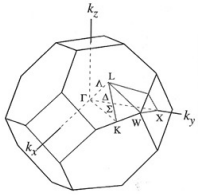


Potential in Crystal

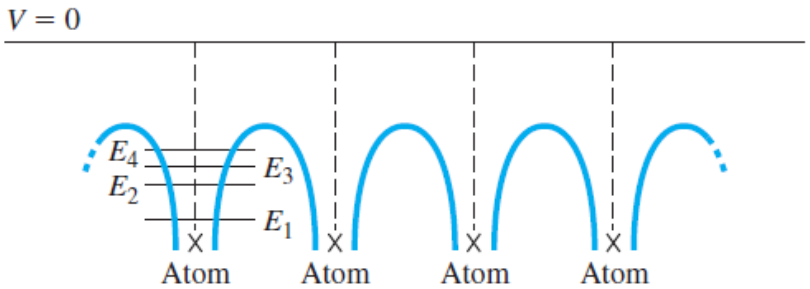




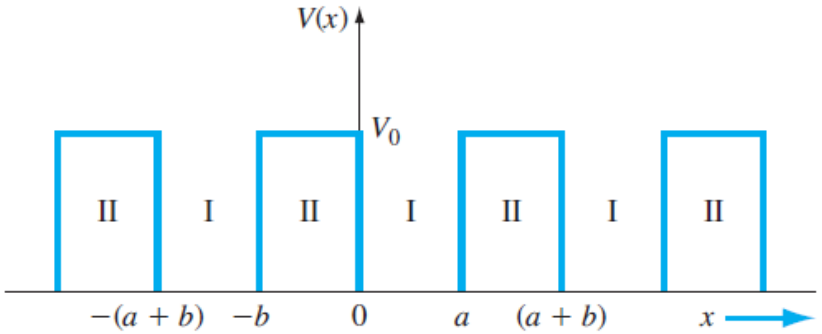




Kronig-Penny Model

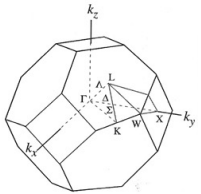


Net potential of 1D single crystal



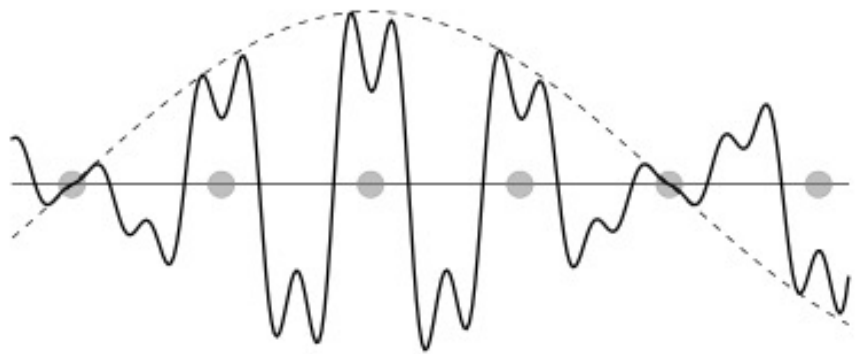
1D periodic potential function of Kronig Penny model

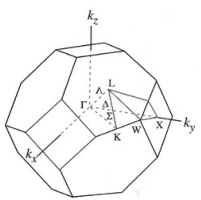
Use Block theorem: $\Psi\big(x+(a+b)\big)=\Psi(x)e^{ik(a+b)}$



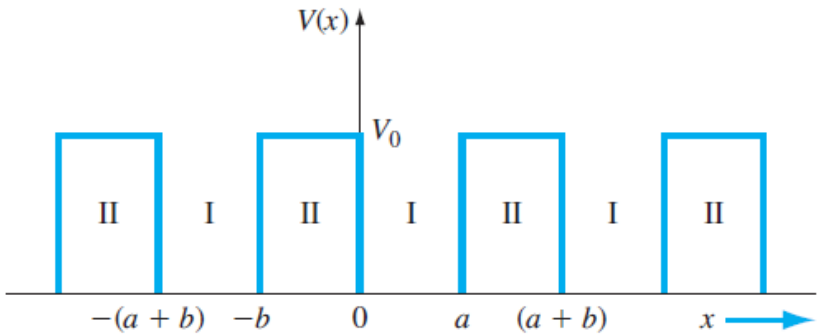
Wavefunction in Crystal

- Bloch Function: $\psi_k(r) = e^{ikr}u_k(r)$
  - e^{ikr} is a plane wave
  - $u_k(r)$modulates the wavefunction according to the periodicity of the lattice
  - $u_k(r) = u_k(r+p)$
  - $\psi_k(r)e^{ikp} = \Psi_k(r+p)$
  - $\Psi(r,t) = \psi_k(r)e^{-j\frac{Et}{\hbar}} = u_k(r)e^{-j(kx-\frac{Et}{\hbar})}$
  - k is also the wave number



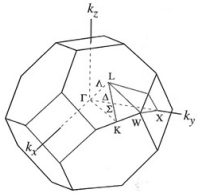


Kronig-Penny Model



- Solve SWE in region I and II
- Use Bloch theorem to match the wavefunction ψ_I and ψ_{II}, $\frac{d\psi_I}{dx}$ and $\frac{d\psi_{II}}{dx}$ at boundaries
- Wavefunction format
  - $0 \le x < a, \psi_a = A_a sin\alpha x + B_a cos\alpha x$
  - $-b < x \le 0, \psi_b = A_b sin\beta x + B_b cos\beta x$
  - $\alpha \equiv \sqrt{\frac{2mE}{\hbar^2}}, \beta \equiv \sqrt{\frac{2m(E-V_0)}{\hbar^2}}$

$$\begin{cases} \psi_b(0^-) = \psi_a(0^+) & P = a+b \\ \psi_b'(0^-) = \psi_a'(0^+) \\ \psi_a(a^-) = \psi_b(a^+) = \psi_b(-b+P) = \psi_b(-b^+)\cdot e^{ikP} \\ \psi_a'(a^-) = \psi_b'(-b^+)\cdot e^{ikP} \end{cases} \Rightarrow \begin{cases} B_a = B_b \\ \alpha A_a = \beta A_b \\ A_a \cdot Sin\alpha a + B_a \cdot Cos\alpha a = [-A_b \cdot Sin\beta b + B_b Cos\beta b]\, e^{ikP} \\ A_a Cos\alpha a - B_a \cdot Sin\alpha a = [A_b\beta \cdot Cos\beta b + B_b\beta \cdot Sin\beta b]\, e^{ikP} \end{cases}$$



Kronig-Penny Model

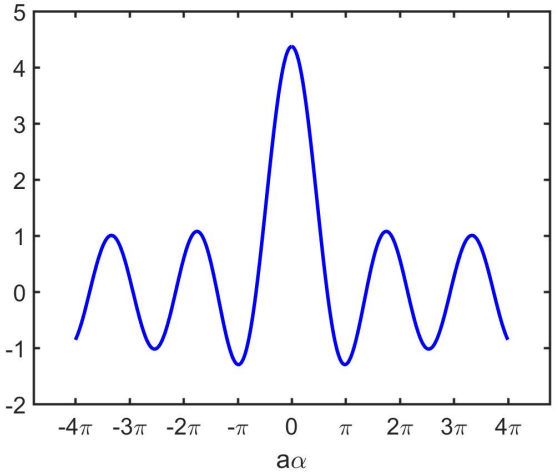
- 4 boundary condition (4 equations)
- Det of coefficients=0 for any solution

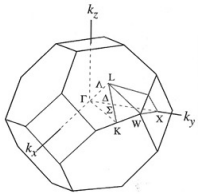
$$\frac{-(\alpha^2+\beta^2)}{2\alpha\beta}(\sin\alpha a)(\sin\beta b)+(\cos\alpha a)(\cos\beta b)=\cos k(a+b)$$

$\alpha, \beta \longmapsto E, k.$

$$\begin{bmatrix} M(E,k) \end{bmatrix}\begin{bmatrix} A_a \\ A_b \\ B_a \\ B_b \end{bmatrix} = 0$$

$\Rightarrow \det(M) = 0$

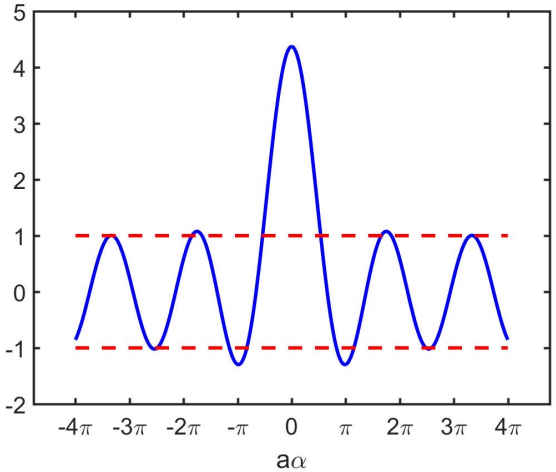


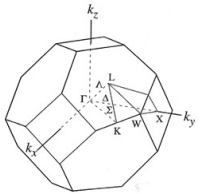


Kronig-Penny Model

- 4 boundary condition (4 equations)
- Det of coefficients=0 for any solution

$$\frac{-(\alpha^2+\beta^2)}{2\alpha\beta}(\sin\alpha a)(\sin\beta b)+(\cos\alpha a)(\cos\beta b)=\cos k(a+b)$$

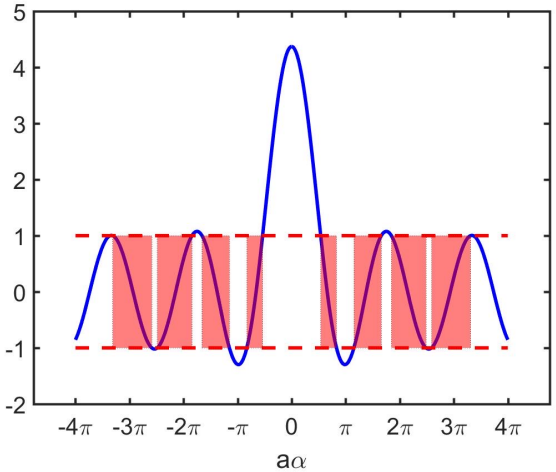


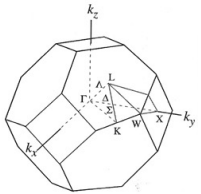


Kronig-Penny Model

- 4 boundary condition (4 equations)
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$$\frac{-(\alpha^2+\beta^2)}{2\alpha\beta}(\sin\alpha a)(\sin\beta b)+(\cos\alpha a)(\cos\beta b)=\cos k(a+b)$$



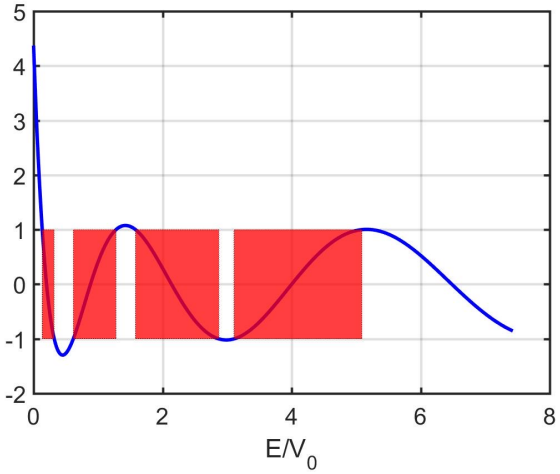


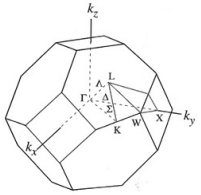
Kronig-Penny Model

- 4 boundary condition (4 equations)
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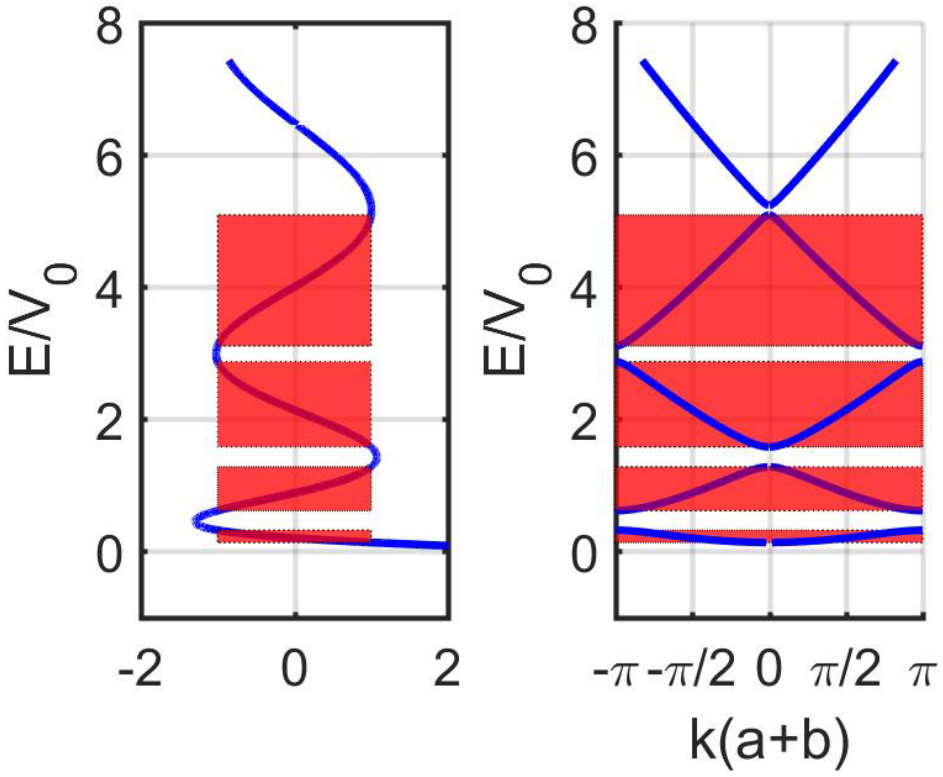
$$\frac{-(\alpha^2+\beta^2)}{2\alpha\beta}(\sin\alpha a)(\sin\beta b)+(\cos\alpha a)(\cos\beta b)=\cos k(a+b)$$

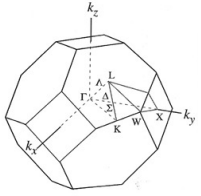
- What k means?



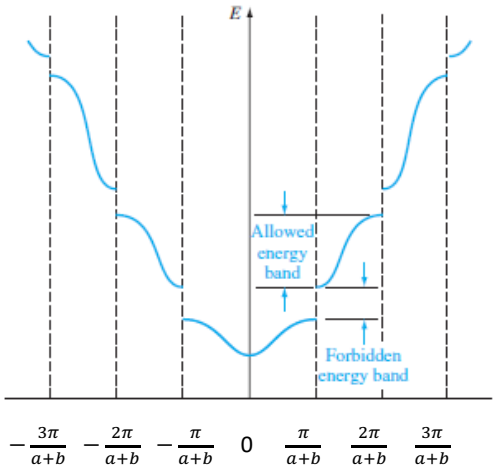


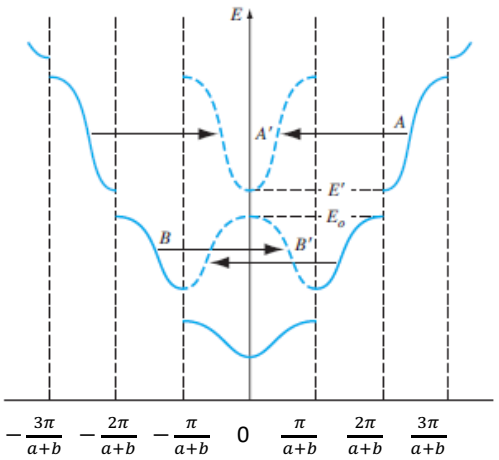
Energy Band

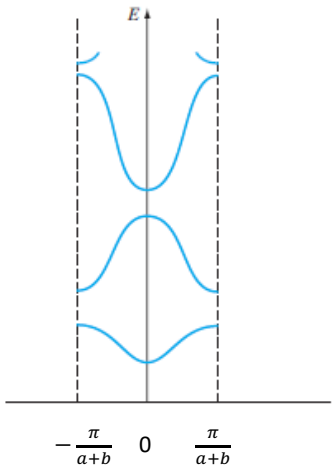




E versus k diagram

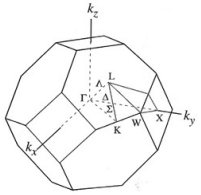






When k change by $\frac{2\pi}{a+b}$, $\cos\left((\mathrm{k}+\frac{2\pi}{a+b})(\mathrm{a}+\mathrm{b})\right)=\cos(k(a+b)+2\pi)=\cos(k(a+b))$

Mathematically, there is no difference when k change by $\frac{2\pi}{a+b}$



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$\varphi(x+p) = \varphi(x) e^{ikp}$

$\varphi(x+Np) = \varphi(x) \cdot e^{ikNp}$

periodic Boundary condition

$\varphi(x+Np) = \varphi(x)$

$e^{iNkp} = 1 \quad \Rightarrow Nkp = 2\pi n$

$k = \frac{2\pi n}{Np}$

$Ae^{j(kx-\omega t)} + Be^{j((k+\Delta k)x-(\omega+\Delta\omega)t)}$

$= Ae^{j(kx-\omega t)} \cdot \left[1 + \frac{B}{A} e^{j(\Delta k x - \Delta\omega t)}\right]$

$v_{group} = \frac{\partial\omega}{\partial k} = \frac{1}{\hbar} \cdot \frac{\partial E}{\partial k}$



E versus k diagram

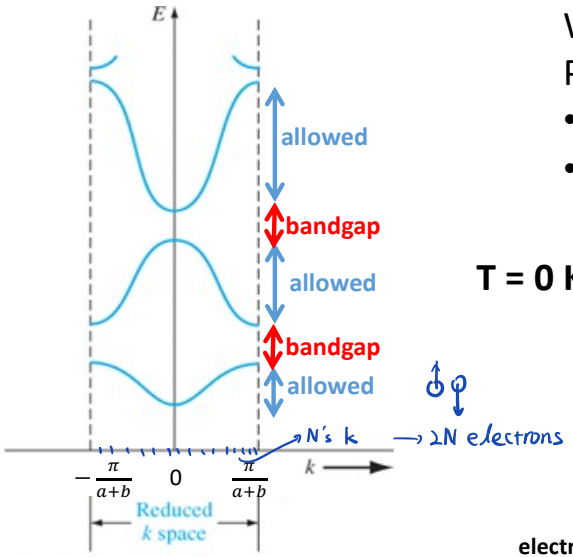


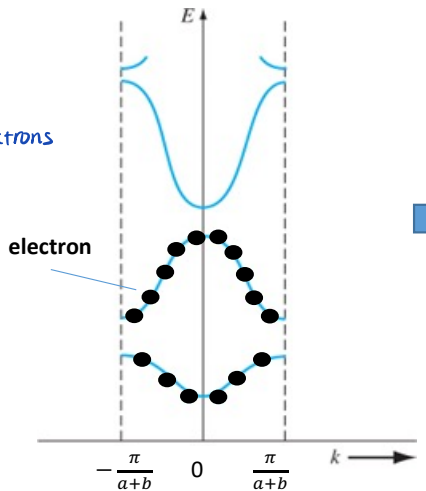
Figure 3.11 | The E versus k diagram in the reduced-zone representation.

Where are the electrons?

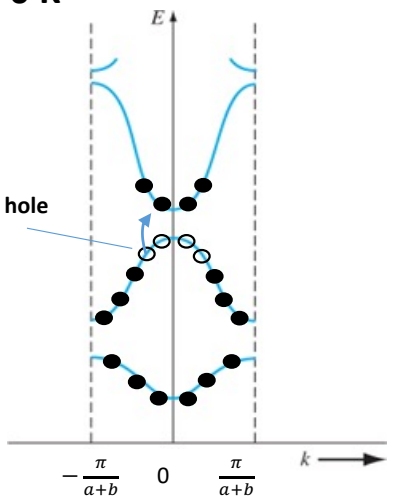
Pauli Exclusion Principle

- No two electrons can have the same quantum state.
- T=0K: fill energy level with electron from bottom to the top until we run out of electron.

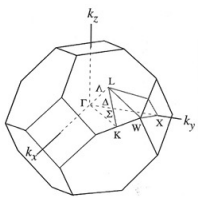
T = 0 K



T > 0 K

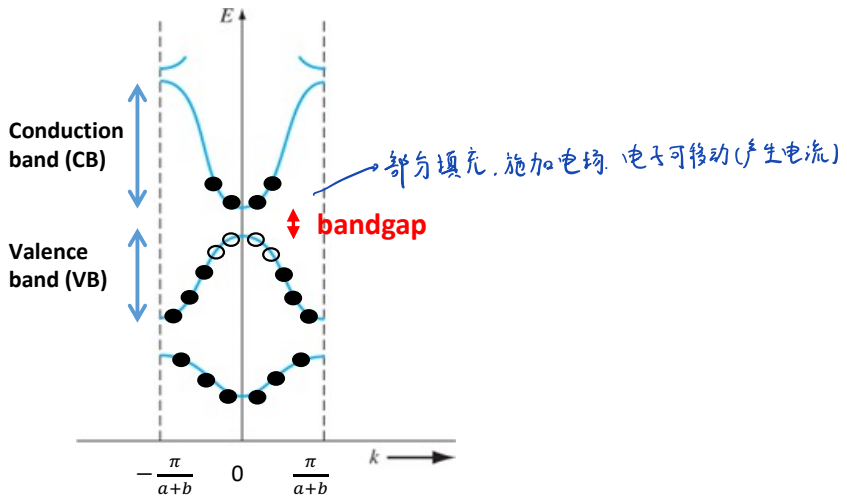


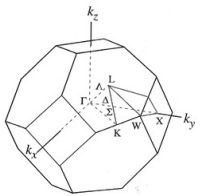
Some of the electrons gain enough thermal energy to overcome the bandgap. They leave a hole (empty electron state) behind.



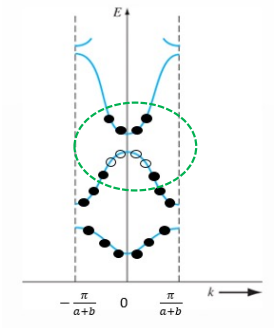
Partly Filled Energy Bands

- We are only interested in partly filled energy bands . Completely filled or completely empty bands do not contribute to conductivity.

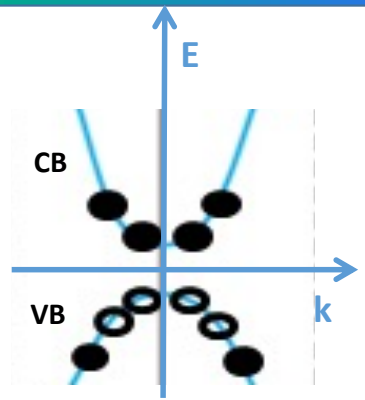


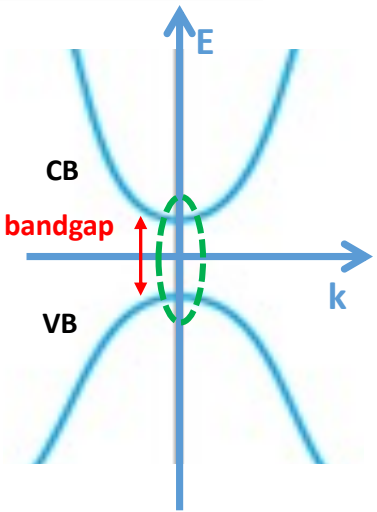


Band Edge

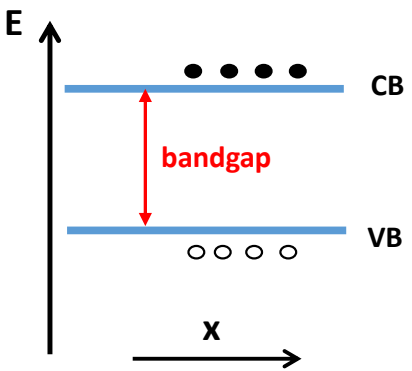


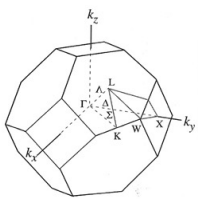
And, we are mostly interested in lowest part of the CB and the topmost part of the VB. Here we find most electrons in the CB and most holes in the VB.



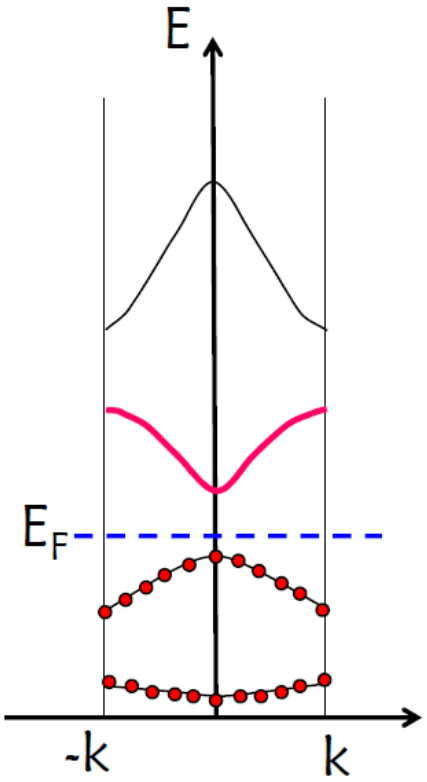


Later, when we connect different types of semiconductors and build devices, we will only plot the bottom of the CB and the top of the VB as function of position x.





Electron and Hole Fluxes: Filled/Empty Bands



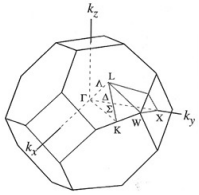
$$v = \frac{\partial \omega}{\partial k} = \frac{1}{\hbar}\frac{\partial E}{\partial k}$$

$$J_3 = -\frac{q}{L}\sum_{i(filled)} \upsilon_i = 0$$

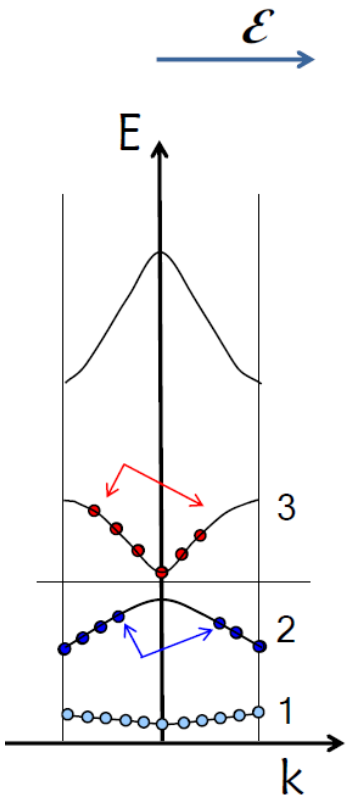
$$J_2 = -\frac{q}{L}\sum_{i(filled)} \upsilon_i = -\frac{q}{L}\sum_{0}^{k_{max}} \upsilon_i - \frac{q}{L}\sum_{-k_{min}}^{0} -|\upsilon_i| = 0$$

Filled and empty bands carry no current !

(For every positive J_2 or J_3 component, there is an equal negative one!)



Electron and Hole Fluxes: Partially Filled Bands



$$J_3 = -\frac{q}{L}\sum_{i(filled)} \upsilon_i \neq 0$$

$$J_2 = -\frac{q}{L}\sum_{i(filled)} \upsilon_i = -\frac{q}{L}\sum_{all} \upsilon_i + \frac{q}{L}\sum_{i(empty)} |\upsilon_i|$$

$$= \frac{q}{L}\sum_{i(empty)} |\upsilon_i|$$

$\upsilon_g = \frac{1}{\hbar}\cdot\frac{\partial E}{\partial k} \xrightarrow{F} F\cdot\Delta x = \Delta E$

$F\cdot\upsilon_g\cdot\Delta t = \Delta E$

$F\cdot\frac{1}{\hbar}\cdot\frac{\partial E}{\partial k}\cdot\Delta t = \Delta E$

$\frac{\partial k}{\partial t} = \frac{F}{\hbar}$

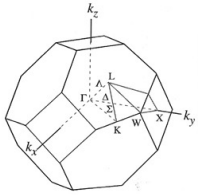
$a = \frac{d\upsilon_g}{dt} = \frac{\partial \upsilon_g}{\partial k}\cdot\frac{\partial k}{\partial t}$

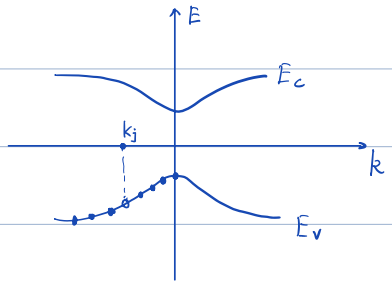
$= \frac{1}{\hbar^2}\cdot\frac{\partial^2 E}{\partial k^2}\cdot F$

$m^* = \frac{\hbar^2}{\frac{\partial^2 E}{\partial k^2}}$

-ve charge moving with –ve mass

+ve charge moving with +ve mass

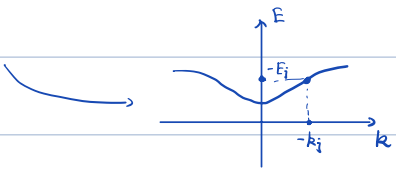




valence band

$E_T = \sum_i E_i - E_j = -E_j$

$k_T = \sum_i k_i - k_j = -k_j$



Metal, Semiconductor, and Insulators

Insulators

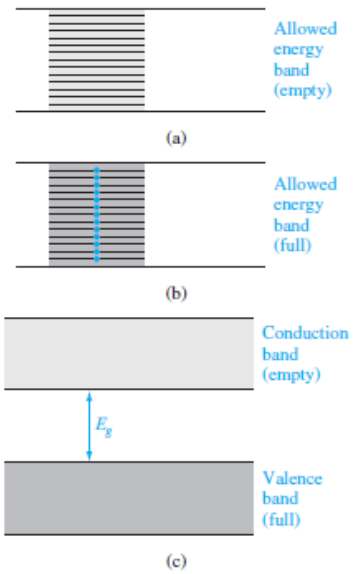


Figure 3.19 | Allowed energy bands showing (a) an empty band, (b) a completely full band, and (c) the bandgap energy between the two allowed bands.

semiconductor

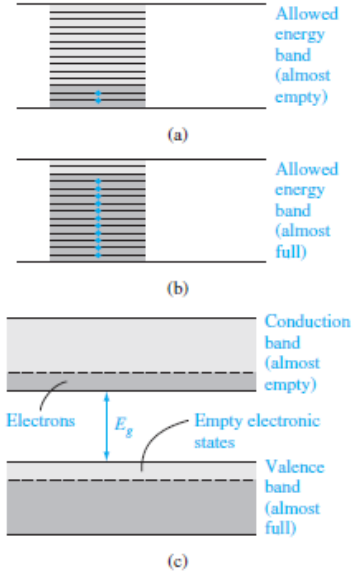


Figure 3.20 | Allowed energy bands showing (a) an almost empty band, (b) an almost full band, and (c) the bandgap energy between the two allowed bands.

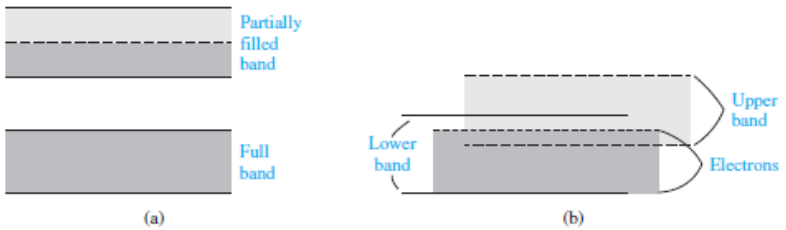
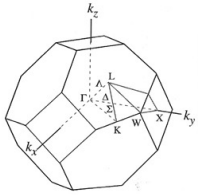


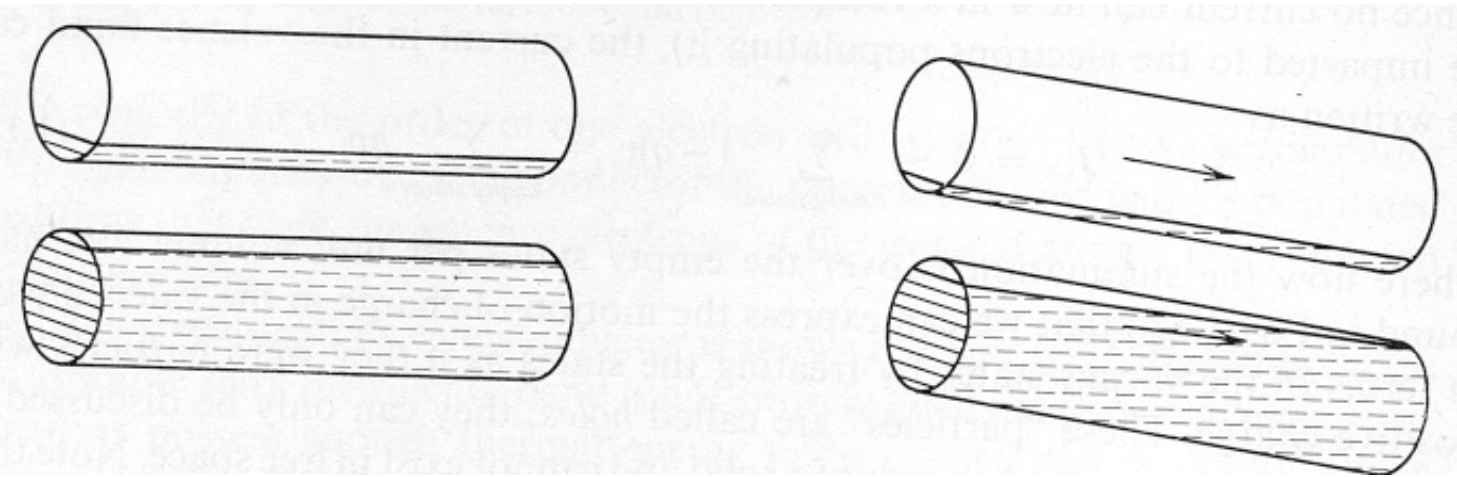
Figure 3.21 | Two possible energy bands of a metal showing (a) a partially filled band and (b) overlapping allowed energy bands.

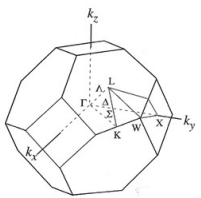
Metal



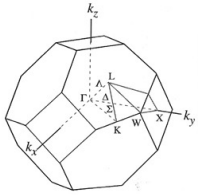
Bottle Flow Analogy

- Water bottle flow analogy (empty vs. full)
- So, what is a hole then?





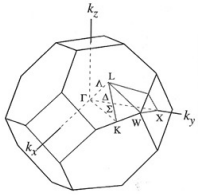
Effective Mass

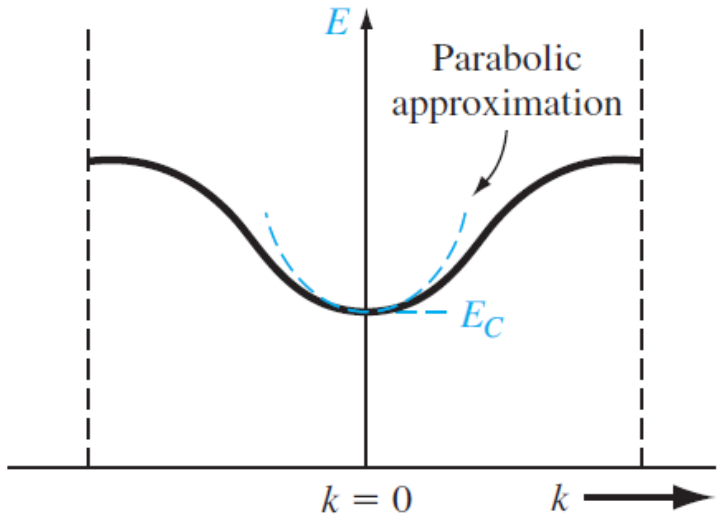


Effective Electron Mass

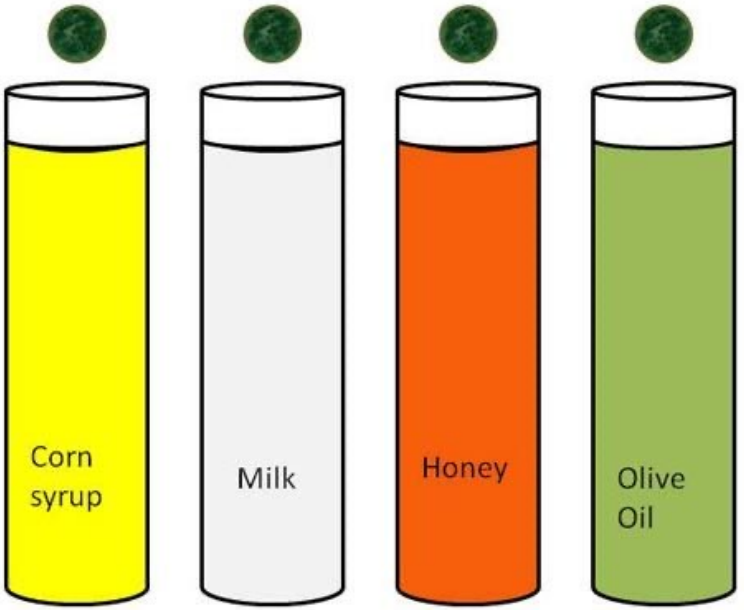
- The movement of an electron in a lattice will be different from that of an electron in free space
  - $F_{total} = F_{ext} + F_{int} = ma$
  - $F_{ext} = m^* a$
- Definition of Effective Mass
  - $\frac{1}{\hbar^2}\frac{d^2E}{dk^2} = \frac{1}{m}$
  - Where does this come from?
- F = m^* · a=-eE

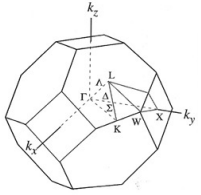
$$a = \frac{-eE}{m^*}$$





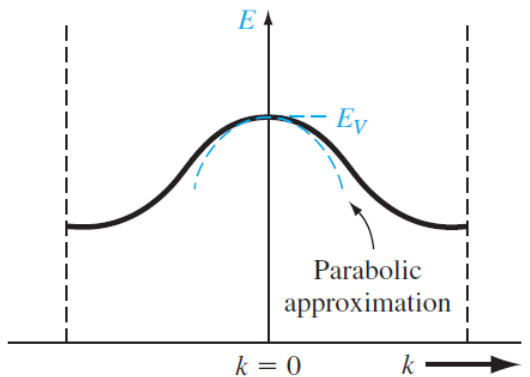
Analogy for Effective Mass Concept

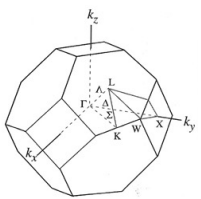




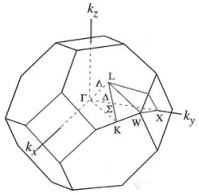
Effective Hole Mass

- Effective Mass in the top of valence band is negative: $\frac{1}{\hbar^2}\frac{d^2E}{dk^2} = \frac{1}{m}$
- F = m^* · a = -eE
  - $a = \frac{-eE}{m^*} = \frac{-eE}{-|m^*|} = \frac{eE}{|m^*|}$
- We now can model this band as having particles with a positive electronic charge and a positive effective mass.
- It is hole.

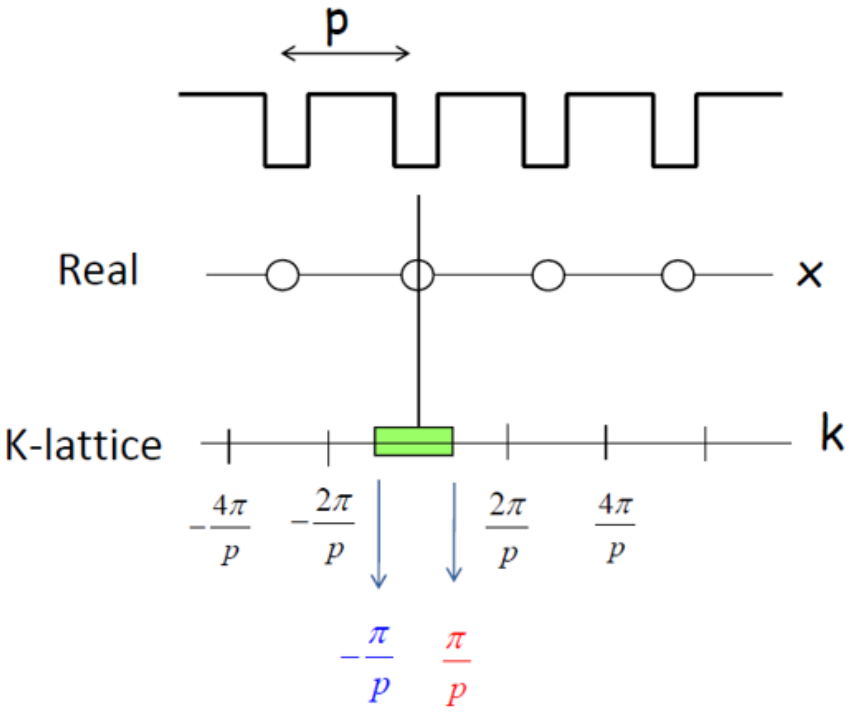


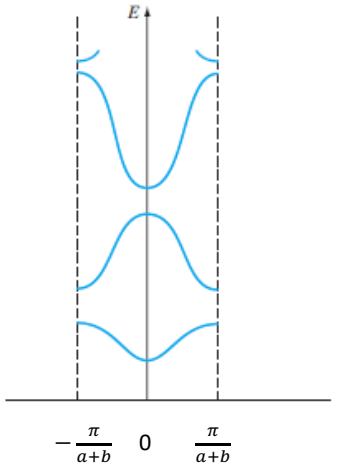


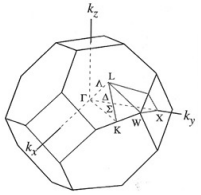
Brillouin Zone



Brillouin Zone





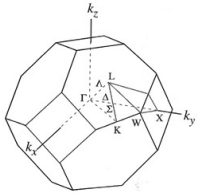


General rules for Brillouin Zone

1) Define reciprocal lattice with the following vectors ….

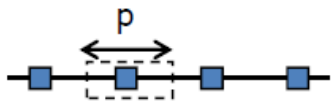
$$k_x = 2\pi \frac{b \times c}{|a \bullet b \times c|} \quad k_y = 2\pi \frac{c \times a}{|a \bullet b \times c|} \quad k_z = 2\pi \frac{a \times b}{|a \bullet b \times c|}$$

2) Use Wigner Seitz algorithm to find the unit cell in the wave-vector (reciprocal) space.



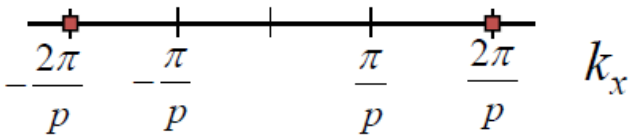
Brillouin Zone for 1-D Solids

Real-space

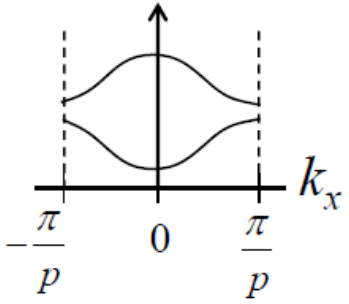


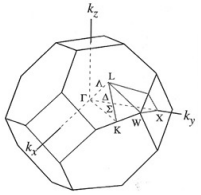
Replacing
(a+b) by p …

1st B-Z



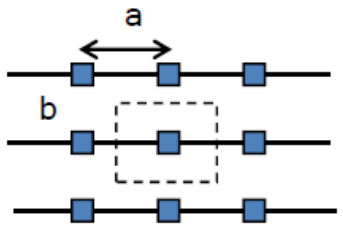
E-k diagram



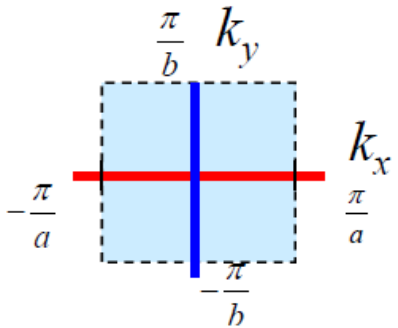


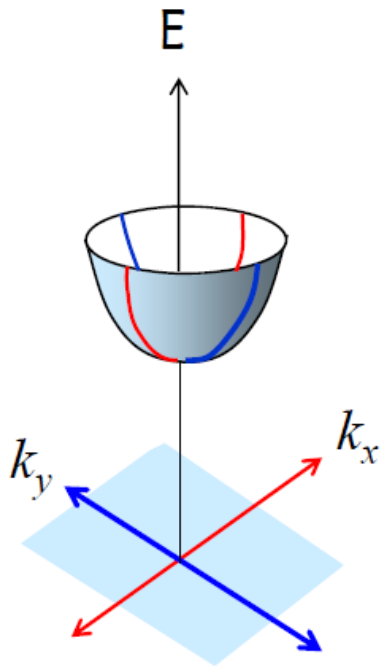
E-k diagram in 2D Solids

Real-space

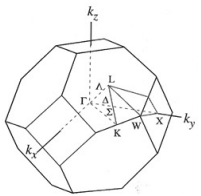


1st B-Z





E-k diagram

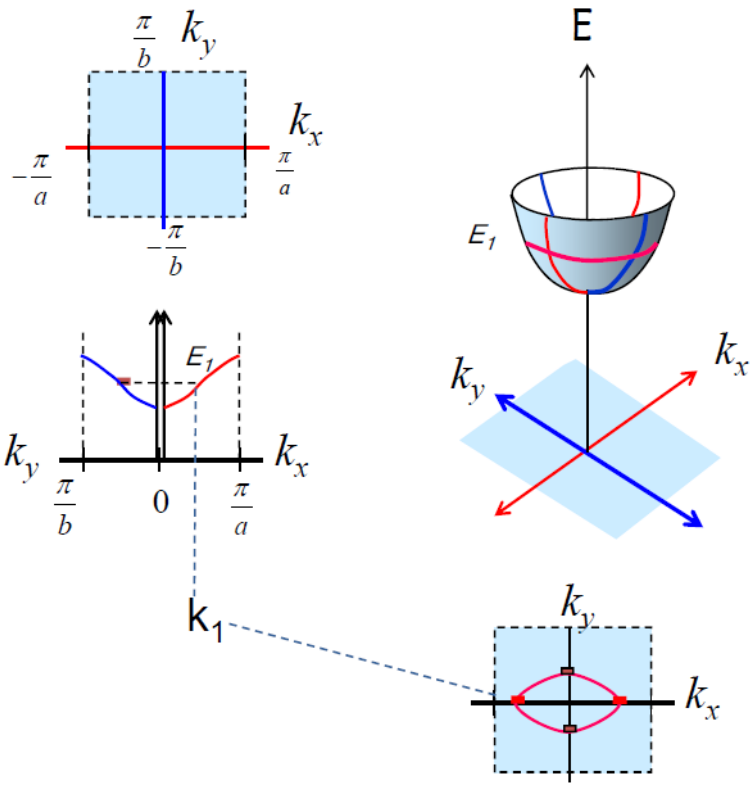


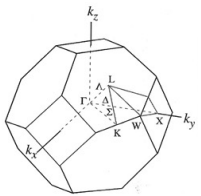
Constant Energy-Surface in 2D

1st B-Z

E-k diagram

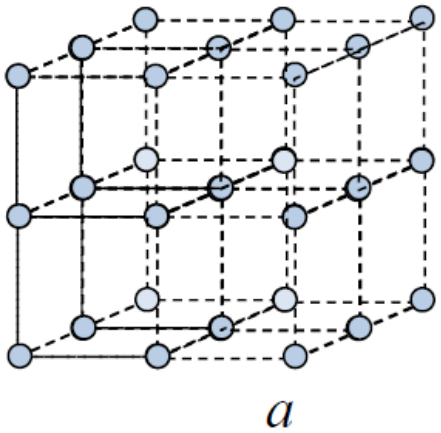
Const. Energy Surface



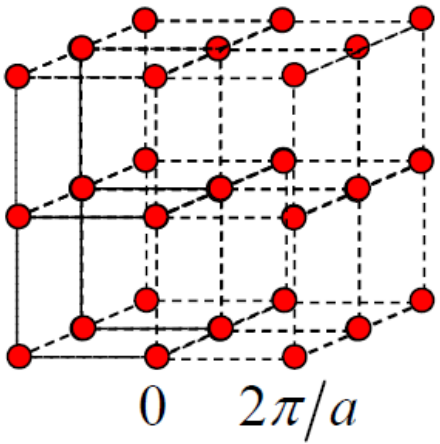


Brillouin Zone in Cubic Lattice

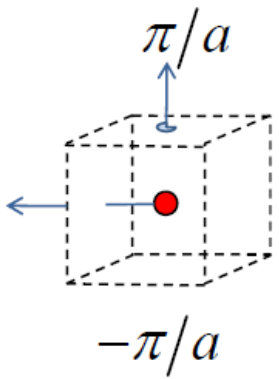
Real Space Cubic Lattice



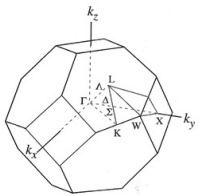
Reciprocal Lattice



Brillouin Zone

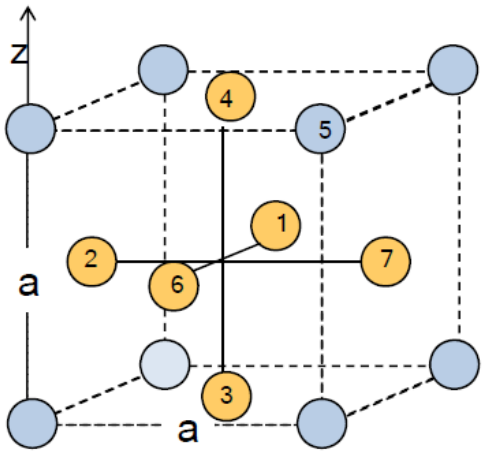


Follow W-S algorithm, but now for reciprocal lattice

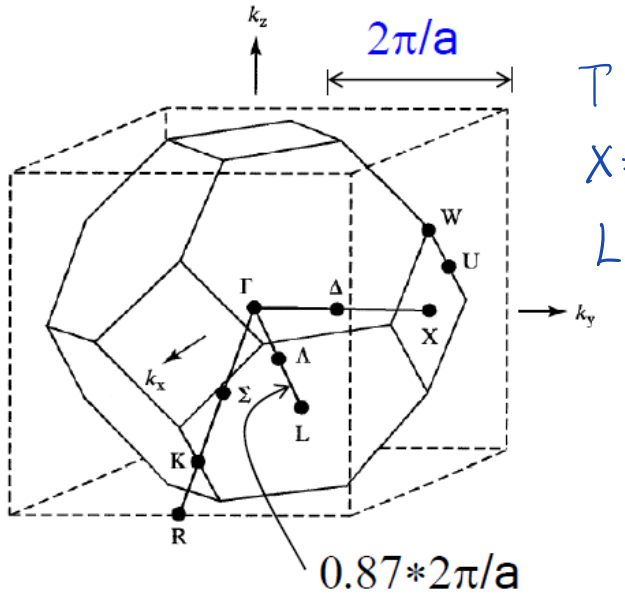


Brillouin Zone in Real FCC Lattices

Real Space FCC (for Si, Ge, GaAs)



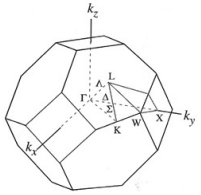
Reciprocal Lattice



Γ : (0, 0, 0)

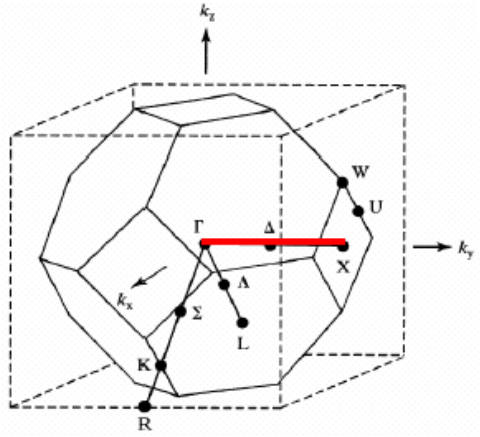
X : $\frac{\pi}{a}$ (1, 0, 0)

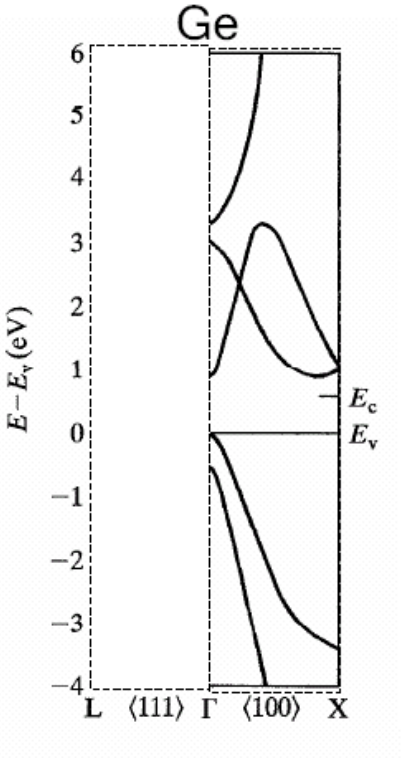
L : $\frac{0.87 \cdot 2\pi}{a}$ (1, 1, 1)

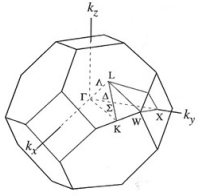


Note unlike cubic lattice, zone edge is not at π/a

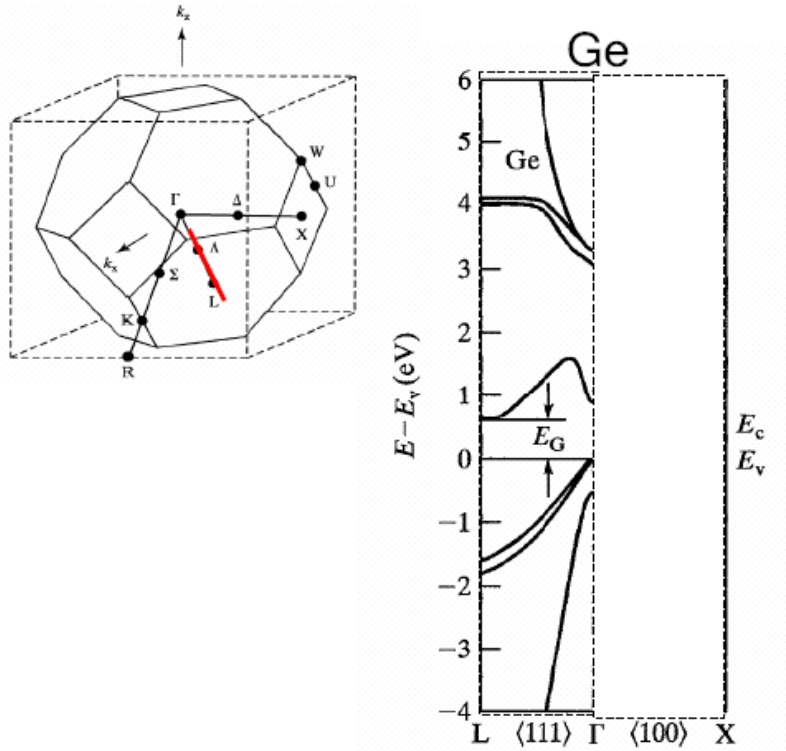
Band Structure along Γ-X direction

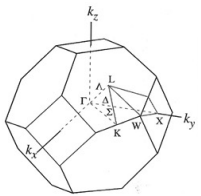




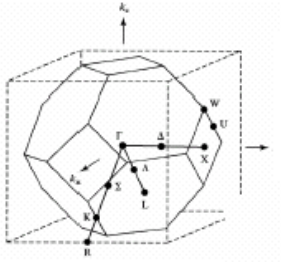


Band Structure along Γ-L direction

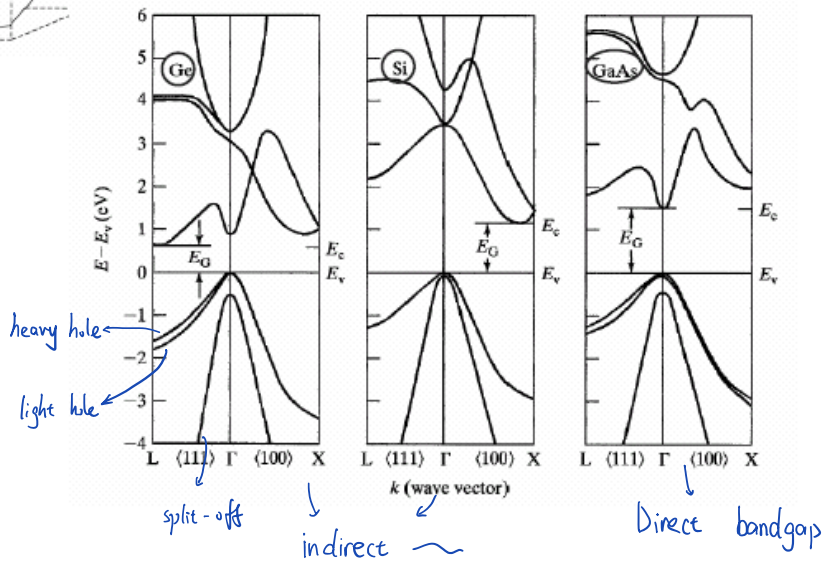


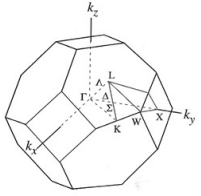


3D Band Structures

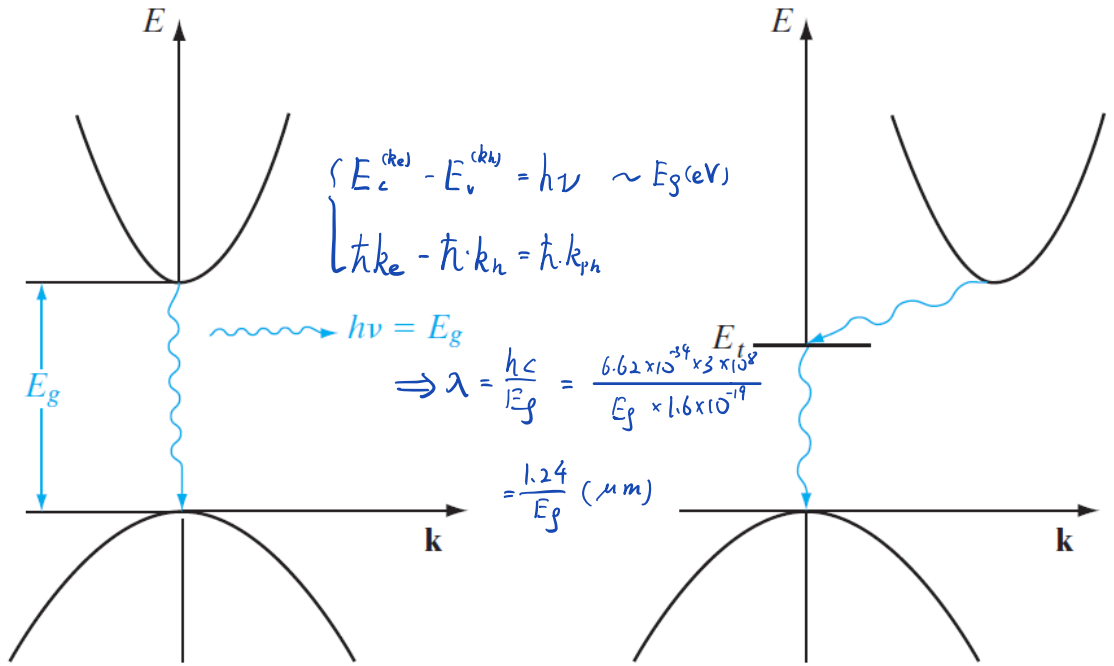


- 3 valence bands (light hole, heavy hole, split-off) valence bands near k=0 is essentially E ~ k^2
- Minima may not be at zone center (Ge: 8 L valley, Si: 6 X valley, and GaAs: Γ valley). May transfer from Γ to L to X.

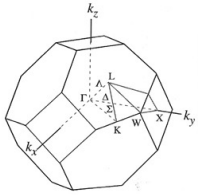




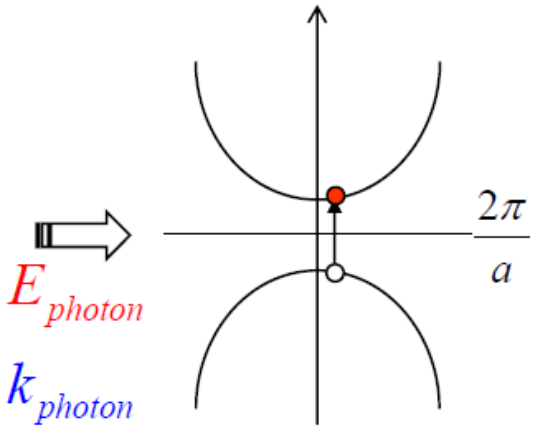
Direct and Indirect Semiconductors



(a) Direct (b) Indirect



Photon Energy and Wavevector

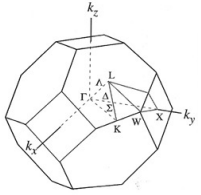


$$E_V + E_{photon} = E_C$$

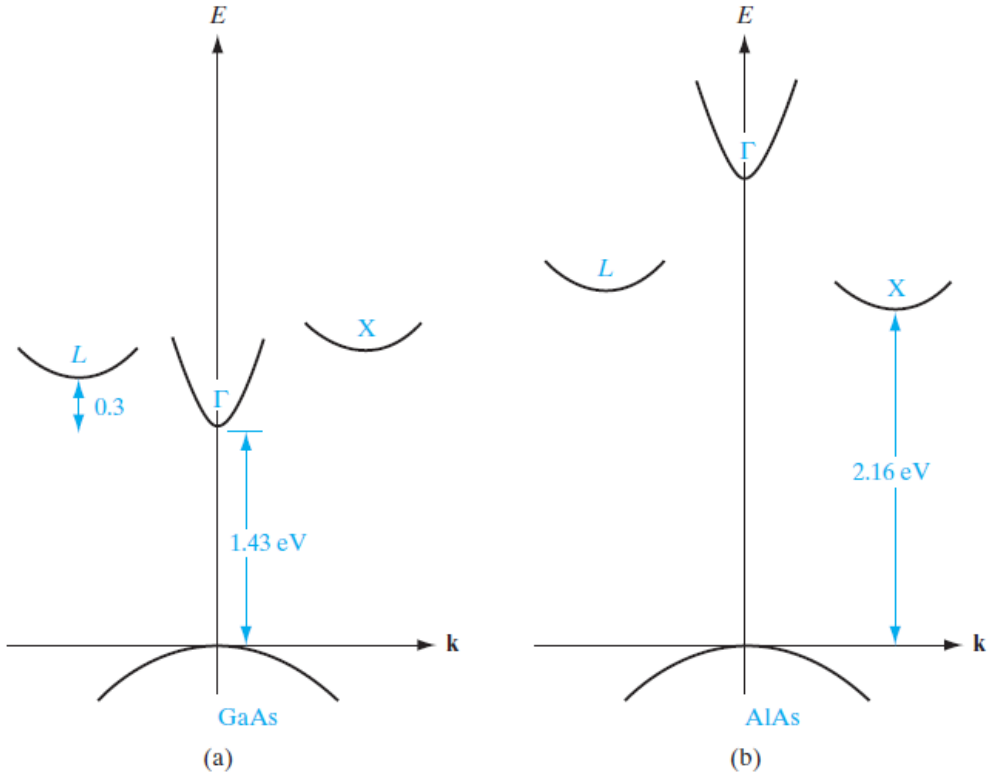
$$\hbar k_V + \hbar k_{photon} = \hbar k_C$$

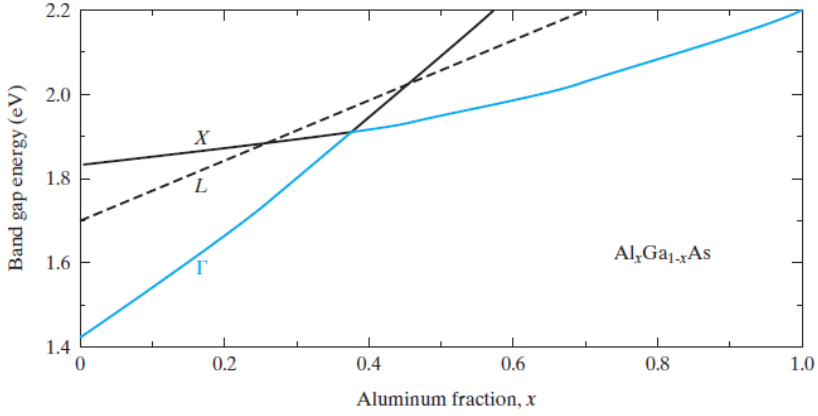
$$k_{photon} = \frac{2\pi}{wavelength}$$

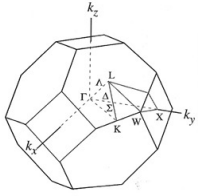
How does this value compare to feature size of Brillouin zone $\frac{2\pi}{a}$?



Variation of Energy Bands with Alloy Composition

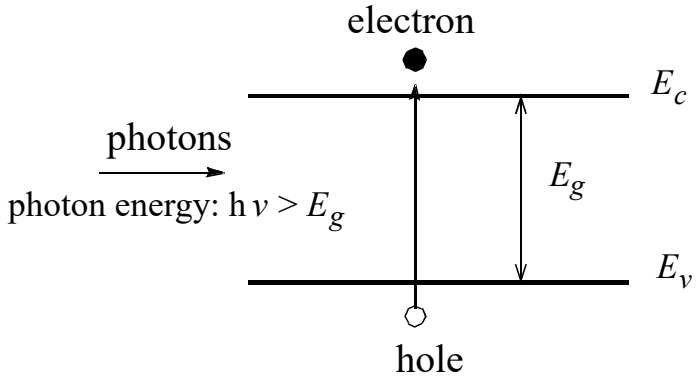






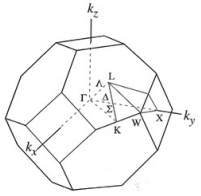
Measuring the Bandgap Energy by Light Absorption

- E_g can be determined from the minimum energy (hu) of photons that are absorbed by the semiconductor.



Bandgap energies of selected semiconductors

Semi-conductor	InSb	Ge	Si	GaAs	GaP	ZnSe	Diamond
Eg (eV)	0.18	0.67	1.12	1.42	2.25	2.7	6



Bandgap and Lattice Constant

